

Ranger – An AI Quiz Game

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Project Summary

Ranger is a fun and challenging web-based number-ordering Player's place numeric trivia answers in the correct position on a number line, combining general knowledge with logic and estimation.

With Classic, Daily, and Legacy modes, the game is designed to be quick to pick up and engaging to return to each day.

How it Works

Players place questions onto a number line between 0 and ∞ . Each placement is validated against neighbouring values. All answers are integer values.

Classic Mode creates a new game each time based off user inputs, **Daily Mode** provides one shared challenge per day.

Legacy Mode allows archived daily challenges to be replayed

Tech Used

Frontend: React, Vite, CSS, dnd-kit

Backend: Python, FastAPI, Uvicorn

Database: PostgreSQL

AI: OpenAI API

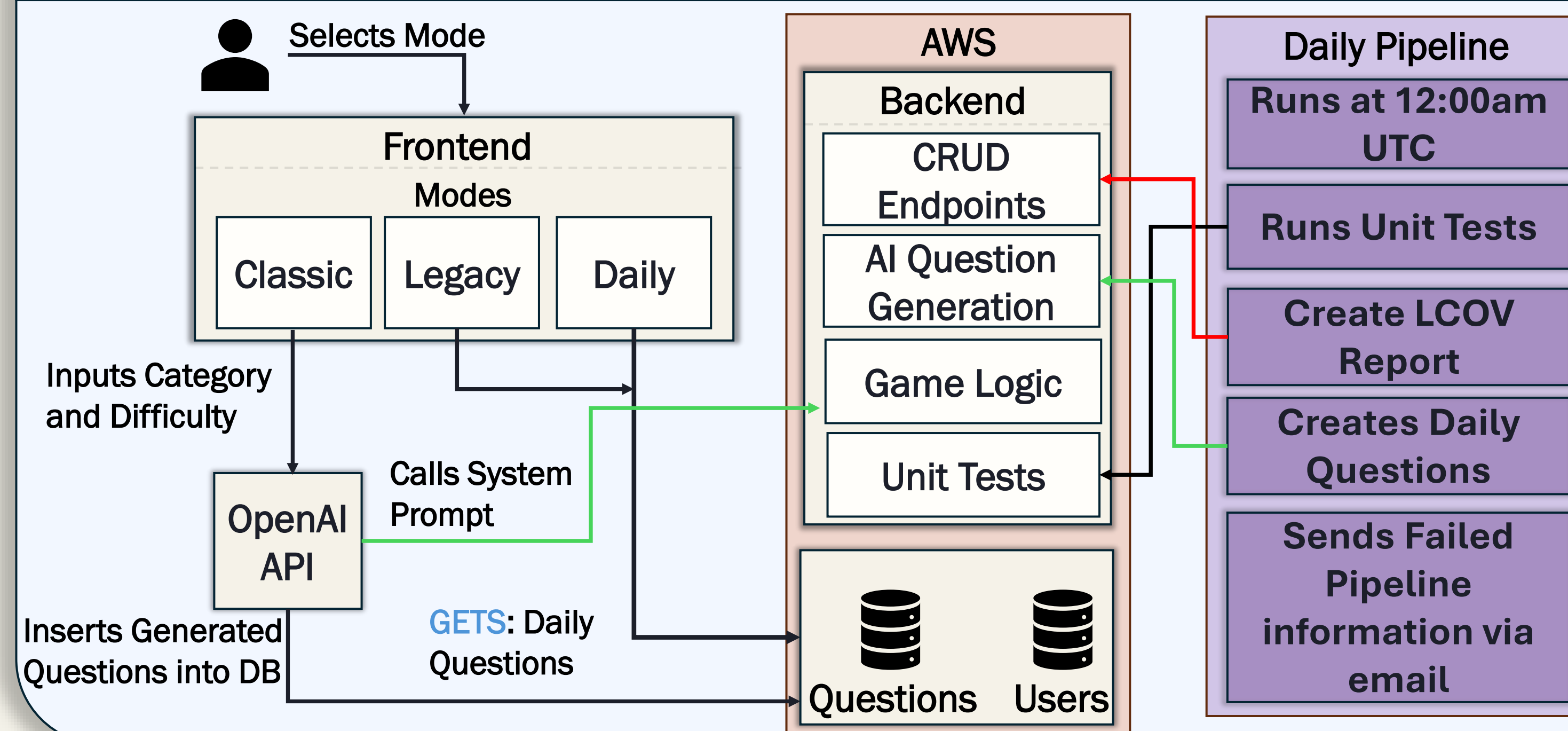
Deployment & DevOps: AWS EC2, Docker, GitHub Actions, Git/GitHub

Testing: pytest, LCOV

Web Application

The screenshot shows the Ranger web application interface. At the top, there's a 'Daily Mode' section with fields for DATE (Sunday 12 April 2026), CATEGORY (USA History), and DIFFICULTY (Hard), along with a 'Reload Daily' button. Below this is a 'Start a new game' section with CATEGORY (Football) and DIFFICULTY (Easy) dropdowns, a 'Start Game' button, and a session ID. To the right is a 'Legacy' calendar for March 2026 with a grid of dates. At the bottom, a 'Current question' box displays a trivia question about the UEFA European Championship, and a number line from 0 to ∞ with three question cards placed between the numbers.

Architecture



CICD & Automation

GitHub Actions is used to automate testing, coverage reporting, and scheduled backend workflows. On every push, the pipeline runs automated tests to validate changes before they are accepted.

A separate nightly workflow triggers the Daily Challenge pipeline, generating and storing the next day's challenge automatically.

Results

This game was tested by my classmates and friends, and feedback was gathered through a post-play survey.

Fun Game, Easy to pick up
★★★★★ Eoin Fitzpatrick
New, refreshing concept. Daily challenges encourage the user to return each day for a new challenge.

AI Mode Allows for Endless Play
★★★★★ Mayzie Potton
Possibilities are endless! And I can play with specific categories I wouldn't find elsewhere.

The Next WORDLE!
★★★★★ David Jayakumar
The AI mode stops the game from getting stale and has the same addictive playstyle as the New York Times WORDLE.

Future Work

The first step for continuing this project would be to deploy it on a full-time server (not just an AWS student server). And to create a mobile app version, allowing for users to play on the go.